

▽ SHAPING THE LAND

The flow of rivers and streams over Earth's surface, causing erosion, is part of the rock cycle and the water cycle, and one of the major processes sculpting surface landscapes.



occurring within Earth's crust and surface continually change crustal rocks from one type into another, in a never-ending succession of events known as the rock cycle. The various components of the rock cycle, which include, for example, volcanism and numerous erosional and depositional processes, are extremely important in shaping Earth's surface landscapes.

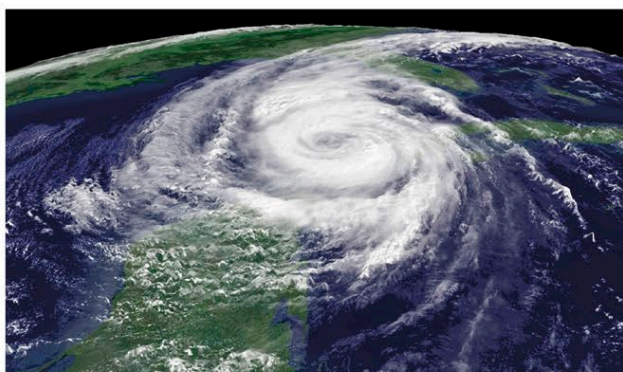
Earth's atmosphere

The atmosphere is part of our planet's overall structure and has several layers. Air circulates only in the lowest layer, which is called the troposphere, and it is only in this layer that weather occurs—taking the form of, for example, winds, precipitation (rain, hail, and snow), and changes in temperature, pressure, and humidity. Weather is driven largely by a combination of energy coming from the Sun and from Earth's rotation. The averaged-out weather in a region over a long period of time is termed that region's climate. Both weather and climate, and the connected processes of the water cycle (which include, for example, precipitation, the flow of water over Earth's land surface in streams and rivers, and the evaporation of water

The mantle transition zone is thought to **contain more water than all the oceans**

from the sea) are other vital factors in shaping landscapes. Changes in Earth's climate, in particular a persistent warming caused by increased levels of carbon dioxide in the atmosphere, are having a widespread effect on surface landscapes.

For example, a general retreat and shrinkage of glaciers is occurring worldwide.



◁ HURRICANE

One of the more extreme forms of weather, a hurricane originates from solar warming of the surface of a tropical ocean. This creates a region of low pressure, with dense clouds above it and winds circulating around it.